

Hyperlocal Courier Dispatch & Tracking Engine

Requirements Table — Lab 1: Requirements Engineering & UML Use-Case Modelling

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall match an outgoing courier request with the nearest active delivery rider within a 3 km radius.	High	Pass: Rider receives dispatch notification with pickup location. Fail: Order assigned to offline rider.	Ensures fast pickup and minimizes idle rider time.	
FR-002	Functional	The system shall allow a Sender Client to create a courier request by entering pickup address, drop-off address, and package details.	High	Pass: A new request record is created and visible in "Pending" status within 2 seconds of submission. Fail: Request is not saved or fields are missing.	Core entry point for the delivery workflow; without it no dispatch can occur.	
FR-003	Functional	The system shall compute an optimized multi-stop route for a rider assigned to more than one active pickup or drop-off.	Medium	Pass: For 2+ assigned stops, the generated route order reduces total distance versus the default sequential order, verified against a routing benchmark. Fail: Route order is unchanged from input order.	Reduces delivery time and fuel/battery cost for riders handling multiple parcels.	
FR-004	Functional	The system shall require the recipient to enter a one-time password (OTP) sent to their registered mobile number before a delivery can be marked complete.	High	Pass: Delivery status changes to "Delivered" only after correct OTP entry. Fail: Status can be marked "Delivered" without OTP entry or with an incorrect OTP.	Prevents fraudulent delivery confirmation and protects both sender and recipient.	
FR-005	Functional	The system shall allow a Delivery Rider to update the delivery status (Accepted, Picked Up, In Transit, Delivered) at each stage of the order lifecycle.	Medium	Pass: Sender Client's tracking view reflects the updated status within 2 seconds of the rider's action. Fail: Status update is not reflected or is delayed beyond 2 seconds.	Keeps the sender informed and enables accurate live tracking.	
NFR-001	Nonfunctional (Performance & Security)	Delivery status and live GPS telemetry updates from riders shall transmit to the sender client with under 2-second latency.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load. Fail: Update latency exceeds 2 seconds in more than 5% of test samples.	Real-time tracking is a core value proposition; stale location data erodes user trust.	
NFR-002	Nonfunctional (Security)	The system shall transmit and store OTP codes using end-to-end encryption, and each OTP shall automatically expire 5 minutes after generation.	High	Pass: Network capture shows OTP payloads encrypted in transit; an expired OTP is rejected by the system. Fail: OTP is observable in plaintext or is accepted after 5 minutes.	Protects the delivery-confirmation mechanism from interception or replay attacks.	